while you might choose the side door on your box; or I might choose the front door and you might choose the top door; and so on."

"Yes, of course," Scully jumps in. "If we call the top door 1, the side door 2, and the front door 3, then the nine possible door combinations are just (1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)."

"Yes, that's right," Mulder continues. "Now here is the point: Of these nine possibilities notice that five door combinations—(1,1), (2,2), (3,3), (1,2), (2,1)—will result in us seeing the spheres in our boxes flash the same color. The first three door combinations are the ones in which we happen to choose the same door, and as we know, that *always* results in our seeing the same color. The other two door combinations, (1,2) and (2,1), result in the same color because the program dictates that the spheres will flash the same color—blue—if either door 1 or door 2 is opened. Now, since 5 is more than half of 9, this means that for more than half—more than 50 percent—of the possible combination of doors that we might select to open, the spheres will flash the same color."

"But wait," Scully protests. "That's just one example of a particular program: *blue, blue, red.* In my explanation, I proposed that differently numbered boxes can and generally will have different programs."

"Actually, that doesn't matter. The conclusion holds for all of the possible programs. You see, my reasoning with the *blue, blue, red* program only relied on the fact that two of the colors in the program are the same, and so an identical conclusion follows for any program: *red, red, blue,* or *red, blue, red,* and so on. Any program has to have at least two colors the same; the only programs that are really different are those in which all three colors are the same—*red, red, red* and *blue, blue, blue.* But for boxes with either of these programs, we'll get the same color to flash regardless of which doors we happen to open, and so the overall fraction on which we should agree will only increase. So, if your explanation is right and the boxes operate according to programs—even with programs that vary from one numbered box to another—we must agree on the color we see *more* than 50 percent of the time."

That's the argument. The hard part is now over. The bottom line is that there *is* a test to determine whether Scully is correct and each sphere